

PLC-CECT Dataset Reference

Primary Liver Cancer — Contrast-Enhanced CT Imaging Dataset

278 Liver Cancer Cases	83 Non-Cancer Controls	361 Total Subjects
50 000+ Annotated Slices	4 CT Phases per Patient	3 Cancer Subtypes

Imaging Protocol & Data Structure

Phase 1 Plain / Non-contrast	Phase 2 Arterial (~25s)	Phase 3 Venous (~60s)	Phase 4 Delayed (~3 min)
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Format	NIFTI (.nii.gz), 3D volumes	Scanner	Single institution, standardized protocol
Cancer types	HCC · ICC · cHCC-CCA (+ 83 controls)	Slice thickness	1–5 mm axial reconstruction

SOTA Segmentation Benchmarks — Ly et al., arXiv:2510.25522 (Nov 2025)

Model	Params	Dice ↑	HD95 ↓	IoU ↑	Acc ↑	Sen ↑	Spe ↑
UNet	1.8M	0.506	224.71	0.491	0.786	0.532	0.789
EfficientUNet	13.2M	0.601	165.98	0.504	0.838	0.649	0.840
SwinUNet	27.6M	0.617	129.81	0.512	0.874	0.678	0.876
MambaUNet	19.1M	0.646	130.92	0.548	0.875	0.705	0.877
TransformerUNet3+	53.3M	0.656	159.38	0.565	0.843	0.678	0.845
MambaUNet3+	36.1M	0.710	112.90	0.617	0.890	0.724	0.892
ResNetUNet3+	31.1M	0.746	88.08	0.657	0.915	0.768	0.916
ResNetUNet3+ + SE	31.2M	0.747	99.93	0.659	0.903	0.761	0.904
ResNetUNet3+ + ASPP	31.3M	0.735	106.10	0.645	0.897	0.754	0.898
ResNetUNet3+ + CBAM+ASPP	31.4M	0.753	90.32	0.662	0.912	0.761	0.914
ResNetUNet3+ + CBAM ★	31.2M	0.755	77.91	0.662	0.925	0.777	0.926

★SOTA ResNetUNet3+ + CBAM: Dice 0.755, HD95 77.91, IoU 0.662, Acc 0.925, Spe 0.926. ResNet outperforms Transformer/Mamba on this dataset.